

persistent target-force context

Causal inputs
 $F_t, \dot{F}_t, F^*, \mathbf{s}_t$

Force-conditioned
latent dynamics

Learned next-force
prediction

Causal risk signal
 $U_{t+1} = \hat{F}_{t+1}$

latent rollouts

Regime + transient
memberships

weights + budget

EMA actor prior

Regime-adaptive
risk-aware MPPI

Cartesian action
 $\Delta \mathbf{x}_t$

